AUTOMATIC STREET LIGHT USING LDR

IoT Internship Project Report

Submitted to: CodeAlpha

Project Domain: Internet of Things (IoT)

Project Title: Automatic Street Light using Arduino and LDR

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Academic Year: 2026–2027

Submitted Date: 10-11-2026

1. INTRODUCTION

An Automatic Street Light System is an electronic system that automatically controls street lights based on the surrounding light intensity. It uses a Light Dependent Resistor (LDR) to detect the amount of light in the environment.

During daytime, when sufficient sunlight is available, the street light remains OFF. During nighttime or in low-light conditions, the LDR detects darkness and the Arduino automatically switches ON the LED.

This project helps reduce unnecessary electricity consumption and demonstrates the application of automation using Arduino and sensors.

2. OBJECTIVE

The main objectives of this project are:

1. To design an automatic street light system using Arduino.
2. To detect surrounding light intensity using an LDR sensor.
3. To automatically switch ON the LED during darkness.
4. To automatically switch OFF the LED during sufficient brightness.
5. To demonstrate the use of sensors and automation in IoT applications.

3. COMPONENTS REQUIRED

The following components are used in this project:

1. Arduino Uno R3
2. LDR (Light Dependent Resistor)
3. 10kΩ resistor
4. 220Ω resistor
5. LED
6. Breadboard

7. Jumper wires
8. Tinkercad simulation software

4. CIRCUIT DIAGRAM

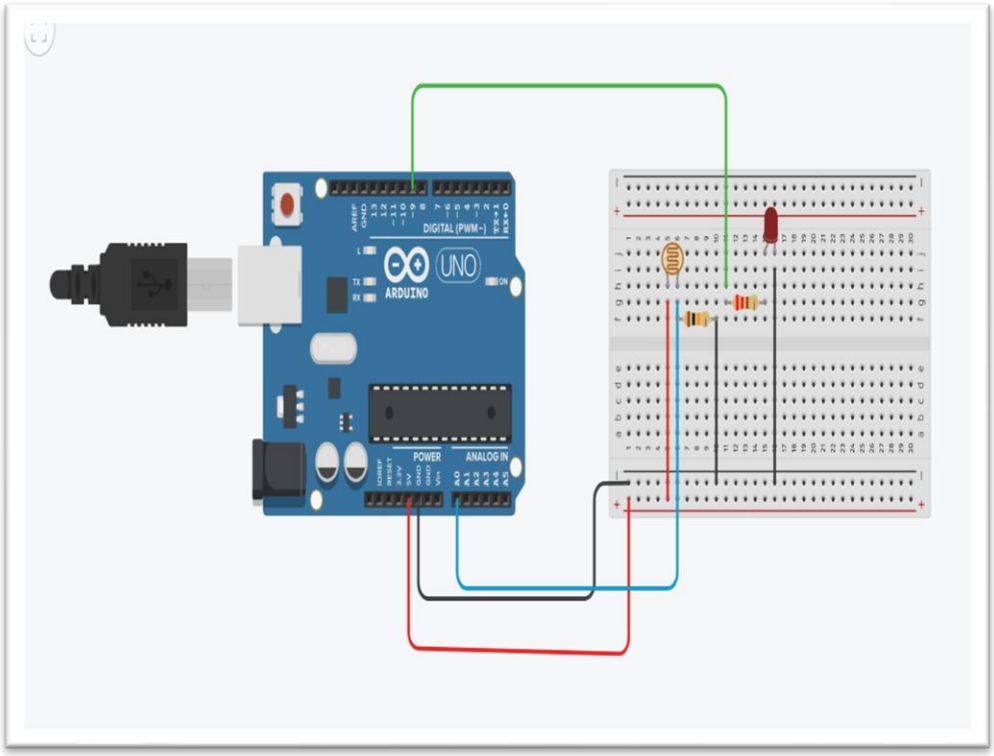


Figure 1: Circuit Diagram of Automatic Street Light using Arduino Uno and LDR

5. CIRCUIT CONNECTIONS

- The LDR is connected to the 5V supply and forms a voltage divider circuit with a 10kΩ resistor.
- The junction between the LDR and resistor is connected to the analog pin A0 of the Arduino.
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- The LED is connected to digital pin D9 through a 220Ω current-limiting resistor. The negative terminal of the LED is connected to GND.
- The Arduino reads the light intensity through the LDR and controls the LED according to the programmed threshold value.

6. WORKING PRINCIPLE

The LDR changes its resistance according to the intensity of light falling on it. When bright light falls on the LDR, its resistance decreases. When darkness occurs, its resistance increases.

The Arduino reads the analog value from the LDR through analog pin A0.

If the LDR reading is below the threshold value of 500, the Arduino switches ON the LED. If the reading is 500 or above, the Arduino switches OFF the LED.

Thus, the system automatically controls the street light according to the surrounding light conditions.

7. ARDUINO CODE

```
int ldrPin = A0;
int ledPin = 9;
int threshold = 500;

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int ldrValue = analogRead(ldrPin);

  Serial.println(ldrValue);

  if (ldrValue < threshold) {
    digitalWrite(ledPin, HIGH);
  }
  else {
    digitalWrite(ledPin, LOW);
  }
```

Code Explanation:

- ldrPin = A0: Reads the LDR sensor value.
- ledPin = 9: Controls the LED.
- threshold = 500: Sets the light detection threshold.
- analogRead(A0): Reads the sensor value.
- digitalWrite(HIGH): Switches ON the LED.
- digitalWrite(LOW): Switches OFF the LED.

8. SIMULATION RESULTS

Darkness Condition

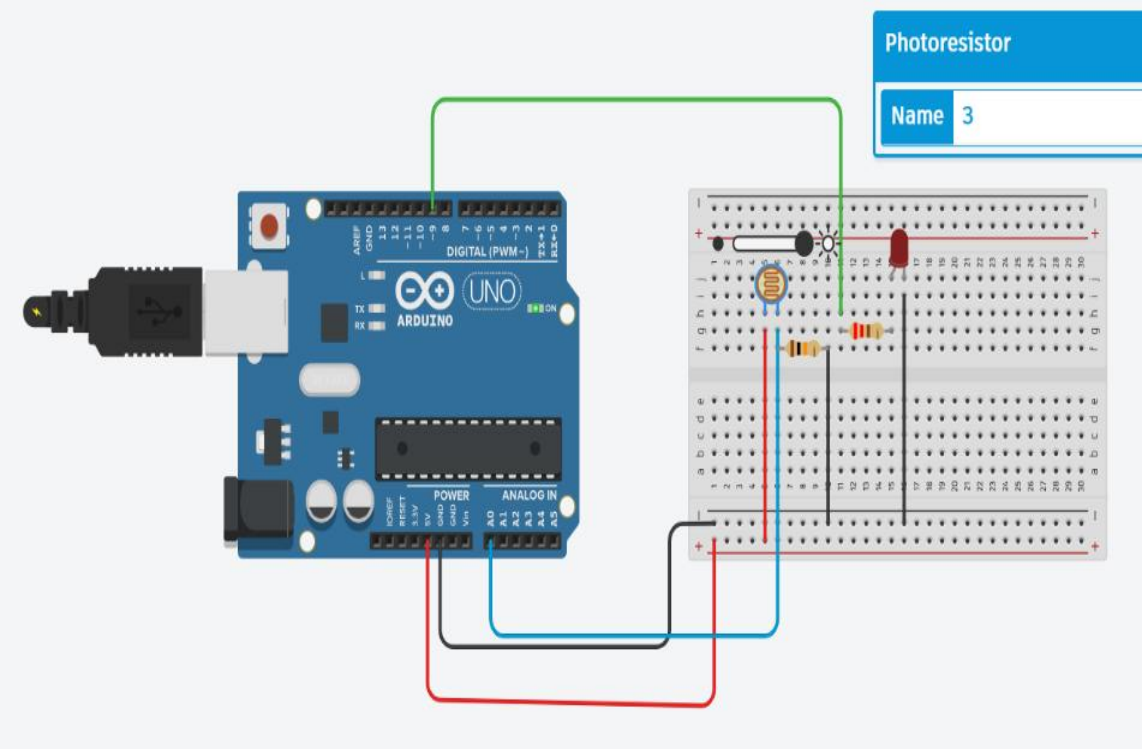


Figure 2: Darkness Condition

Brightness Condition

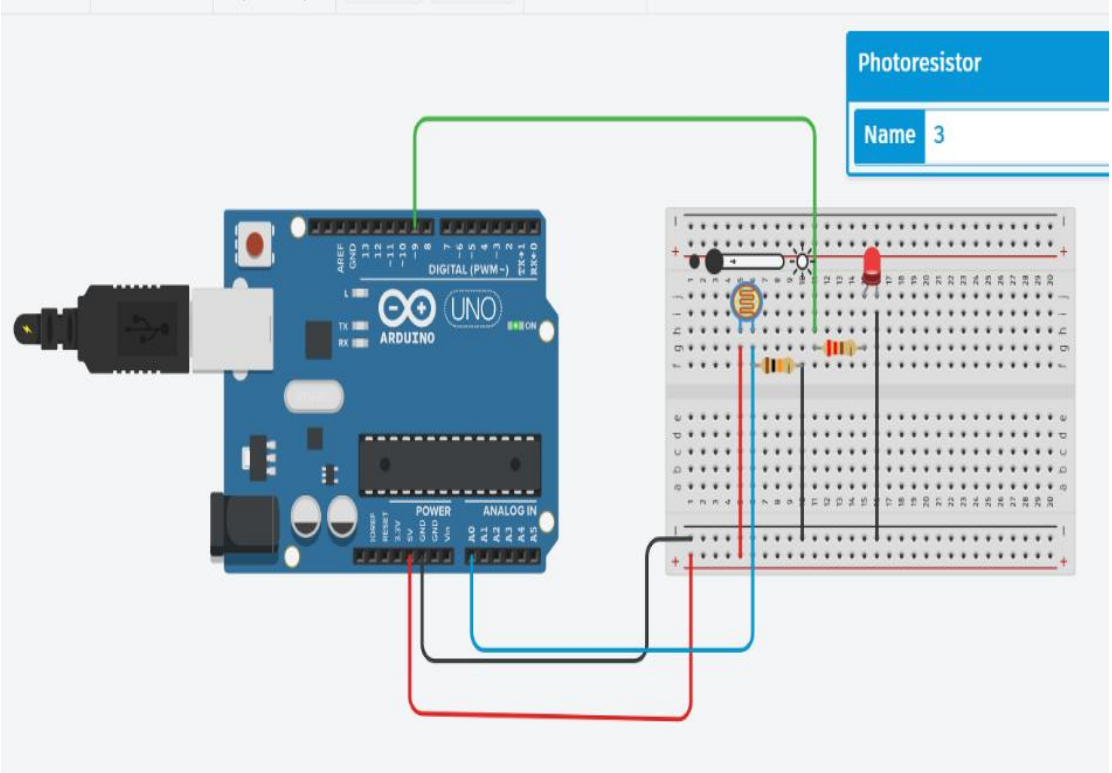


Figure 3: Brightness Condition

9. SIMULATION RESULTS

The project was successfully simulated using Tinkercad.

Test 1: Low Light Condition

When the light intensity was reduced, the LDR reading decreased below the threshold value. The Arduino switched ON the LED automatically.

Test 2: Bright Light Condition

When the light intensity was increased, the LDR reading became higher than the threshold value. The Arduino switched OFF the LED automatically.

The simulation demonstrates the successful operation of the automatic street light system.

10. ADVANTAGES

1. Reduces unnecessary electricity consumption.

2. Automatically controls street lights.
3. Simple and low-cost design.
4. Requires minimum human intervention.
5. Demonstrates sensor-based automation.

11. APPLICATIONS

1. Smart street lighting systems.
2. Residential areas.
3. Parking areas.
4. Gardens and pathways.
5. Smart city infrastructure.

12. CONCLUSION

The Automatic Street Light using LDR project was successfully designed and simulated using Arduino Uno and Tinkercad. The system detects the surrounding light intensity and automatically controls the LED.

The LED switches ON during low-light conditions and switches OFF during bright conditions. This project demonstrates how sensors and microcontrollers can be used to develop energy-efficient automation systems.

The project provides a basic understanding of sensor interfacing, Arduino programming, and practical IoT applications.